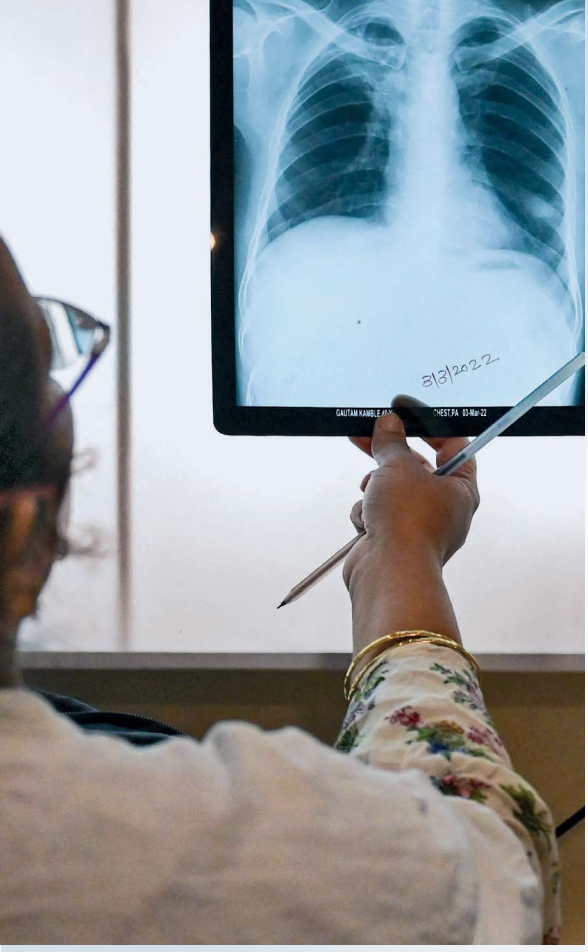


TB TREATMENT AND CHALLENGES

A silent invader spread through the air, TB is caused by a tenacious bacterium that primarily targets the lungs. It typically causes persistent cough, fever, night sweats and weight loss. But if left untreated, it can lead to serious complications such as lung damage, meningitis or infection of other organs

WHAT IS THE PRESENT TB TREATMENT REGIMEN IN INDIA?

- The treatment of newly diagnosed drug-sensitive (DS) TB—which responds to all first-line drugs—comprises oral intake of four drugs (Isoniazid, Rifampicin, Ethambutol and Pyrazinamide) for two months followed by three drugs (Isoniazid, Rifampicin and Ethambutol) for another two months
- The treatment regimen for multidrug-resistant TB (MDR-TB) consists of usually four months of seven drugs (Bedaquiline, Levofloxacin, Clofazimine, Isoniazid, Ethambutol, Pyrazinamide and Ethionamide) followed by five months of four drugs (Levofloxacin, Clofazimine, Pyrazinamide and Ethambutol)
- In about 30 per cent DR-TB patients, Cycloserine and Linezolid are also prescribed; in most cases, treatment lasts up to 1-2 years
- Side effects range from mental disorders, hallucinations, severe body or nerve pain, nausea, and weight loss

WHAT IS THE NEW BPaL REGIMEN?

- In Dec. 2022, the WHO came out with a new regimen for MDR-TB. Known as BPaL, it consists of only three drugs—Bedaquiline, Pretomanid and Linezolid—and reduces the treatment time to six months
- Fewer medicines and lesser duration promise better compliance among patients who otherwise stop taking pills after the acute phase is over. The global treatment success rate is 63 per cent
- Despite widespread availability of the drugs, it took India nearly two years to finally approve its rollout in August

WHY IS CONSISTENT DRUG SUPPLY CRUCIAL?

- TB drugs are heavily subsidised in India, and their procurement is centralised under the National Tuberculosis Elimination Programme
- But twice in less than a year—August 2023 and March 2024—there have been mass-scale reports of drug shortages for weeks
- Incomplete or irregular treatment allows the bacteria to develop resistance to drugs. This drug-resistant strain poses danger to others as well

"Many patients don't return for medicine after the acute phase is over. That [can lead to] drug resistance and community spread of the bacteria," adds Dr Dutta. "The next step is to reduce the length of treatment and the number of pills."

The New Regimen

This is exactly what the BPaL regimen promised. The WHO came out with it in December 2022 to treat MDR and Rifampicin-resistant (RR) TB. This novel six-month regimen comprises just three oral drugs—Bedaquiline, Pretomanid and Linezolid (hence the name BPaL). Bedaquiline, which disrupts energy production in the TB bacteria, has been developed by the American pharma major Johnson & Johnson, and has been an important component of India's DR-TB treatment programme. Pretomanid has been developed by the not-for-profit TB Alliance and stops the bacteria from multiplying by limiting the oxygen supply. Linezolid is an older, generic drug that inhibits protein synthesis, thus halting bacterial growth.

The global treatment success rate of BPaL has been 63 per cent, while the WHO has reported a success rate of nearly 90 per cent in its clinical trials. However, even though the Indian